

MID-TERM EXAMINATION — SPRING 2024

GENERAL SCIENCE & MATHEMATICS

Duration: 2 Hours

Max Marks: 100

Instructions:

1. Read all questions carefully before answering.
2. Write all answers in the provided answer booklet.
3. Scientific calculators are permitted for Section A.

SECTION A: MATHEMATICS & LOGIC

1. (a) Solve for x : $3x^2 - 14x + 8 = 0$ [5 Marks]
(b) If $f(x) = 2x - 5$ and $g(x) = x^2 + 1$, find $g(f(3))$. [5 Marks]
2. A farmer has a circular field with a radius of 70 meters. A path 7 meters wide is built inside the boundary. Calculate the area of the path. (Take $\pi = 3.14$) [8 Marks]
3. A ladder 10m long leans against a vertical wall. The base of the ladder is 6m away from the wall.
(a) How high up the wall does the ladder reach? [5 Marks]
(b) If the base slides 2m further from the wall, how much does the top slide down? [5 Marks]

SECTION B: BIOLOGY & ECOLOGY

4. Define 'Photosynthesis'. Write the balanced chemical equation representing the overall process. [3 Marks]
5. List three key differences between Mitosis and Meiosis. [5 Marks]
6. A food chain consists of: $Grass \rightarrow Rabbit \rightarrow Snake \rightarrow Hawk$
(a) Identify the Primary Consumer and the Tertiary Consumer. [4 Marks]
(b) What happens to the Snake population if all Rabbits die of disease? [3 Marks]

SECTION C: PHYSICS & CHEMISTRY

7. State Newton's Three Laws of Motion with real-world examples for each. [10 Marks]
8. Draw a simple labeled diagram of a standard dry cell (battery), clearly identifying the cathode, anode, and electrolyte. [10 Marks]

SECTION D: KNOWLEDGE, THEORY & COMPUTING

9. (a) Who was the first woman to win a Nobel Prize, and in what field? [3 Marks]
(b) Which planet in our solar system is known as the 'Red Planet'? [2 Marks]

10. (a) Distinguish between ROM and RAM in a computer system. [5 Marks]

(b) Write a simple Python function to find the maximum of three numbers. [5 Marks]

[END OF QUESTION PAPER]